

Fig. 7: Near-field (NF) and far-field (FF) regions of antennae

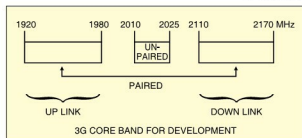


Fig. 8: 2.1GHz band for 3G communications

radiating near-field and radiating far-field—as shown in Fig. 7. (Note that the boundaries shown in the figure are indicative and not well-defined.)

Reactive near-field. It is the region immediately surrounding the antenna wherein the reactive field predominates. Reactive near-field pattern is spherical and decays rapidly with the distance from antenna. Radial limit R_1 of reactive near-field is given by:

$$R_1 = 0.62 \sqrt{D^3/\lambda}$$

where 'D' is the largest dimension of the antenna and ' λ ' is the wavelength. (To be valid, 'D' must be large compared to the wavelength.)

Radiating near-field. This region lies between the reactive near-field region and the far-field region wherein radiation field predominates (in the presence of reactive field). Here the radial field component may be appreciable, while the field pattern is dependent upon the distance from the antenna. Radial limit R_2 of the radiating near-field is given by:

$$R_2 = 2 D^2/\lambda$$

Radiating far-field. The field region beyond radiating near-field is radiating far-field, which theoretically extends to infinity. Here the field pattern is essentially independent of the distance from the antenna. In the far-field region, the field components are transverse and propagate as a plane wave (refer Fig. 5). The radiating far-field decays inversely as its distance from the antenna.

Cellular base station frequency bands

Region-wise, majority of cellphone networks across the world use GSM 900 and GSM 1800MHz bands, including Europe, Asia (including India), Australia, Middle East and Africa, while GSM 850 and GSM 1900MHz bands are used in the United States, Canada, Mexico and most countries of South America.

Additionally, the ITU has identified 1885-2025MHz and 2110-2200MHz bands for 3G development. Of this, the 1920-1980MHz band paired with the 2110-2170MHz band (also known as 2.1GHz band, see Fig. 8) was identified as the core band for development of 3G, and the same has recently been auctioned by the government of India. The 2.1GHz band could make use of

either W-CDMA (wide-band code-division multiple-access) or CDMA-2000 access technology.

Thus changeover to 3G will entail seamless upgradation of 2G/2.5G access technology to 3G access technologies and availability of matching handsets.

Far field of cellphone antennae.

Start of far-field (distance $\geq 2D^2/\lambda$) for cellphone frequency bands of interest, namely, 900, 1800 and 2100 MHz (corresponding to $\lambda=0.333, 0.166$ and 0.143 metres, respectively), can be computed easily from the knowledge of larger dimensions of the antennae; typically, 2.25 metres for 900 MHz and 1.35 metres covering both 1800 and 2100MHz bands (refer the cellphone antennae product details on the website of Kaveri Telecom Products, Bengaluru). The far-field start distances work out to be 30.5 metres for 900MHz antenna and 25.5 metres for 1800 and 21MHz band antennae.

E field, H field and power density in far-field region

From practical viewpoint, for power density measurements, we shall only be concerned with the far-field region. In the far-field region ($>2D^2/\lambda$), the E (electric field intensity) and H (magnetic field intensity) vectors and the direction of propagation are mutually perpendicular to each other; and the quotient of the amplitude of E/H is constant throughout the far-field space. In free space, the ratio of their amplitudes ($E/H = 120\pi$ ohms or 377 ohms) is referred to as characteristic impedance Z_0 of free space. Power density (P_D), i.e., the power per unit area normal to the direction of propagation, is related to the electric and magnetic fields as:

$$P_D = |E| \cdot |H| = E^2/377 = 377 H^2 \text{ watt/m}^2 \dots \text{Eq. (1)}$$

where E is measured in volts (RMS) per metre and H in ampere (RMS) per metre.

Thus by measuring either electric field intensity (V/m) or H-field intensity (A/m) at a far-field location, power density P_D (watts per square metre) can

Definitions of electric and magnetic field strengths

Electric field strength. The standard unit of electric field (E-field) strength is volt per metre (V/m). An E field of 1V/m is represented by a potential difference of 1V existing between two points that are 1 metre apart.

Magnetic field strength. When current flows in a conductor, it is always accompanied by a magnetic field. The strength, or intensity, of this field is proportional to the amount of current and inversely proportional to the distance from the conductor. The unit of magnetic field (H field) is ampere per metre (Am⁻¹).